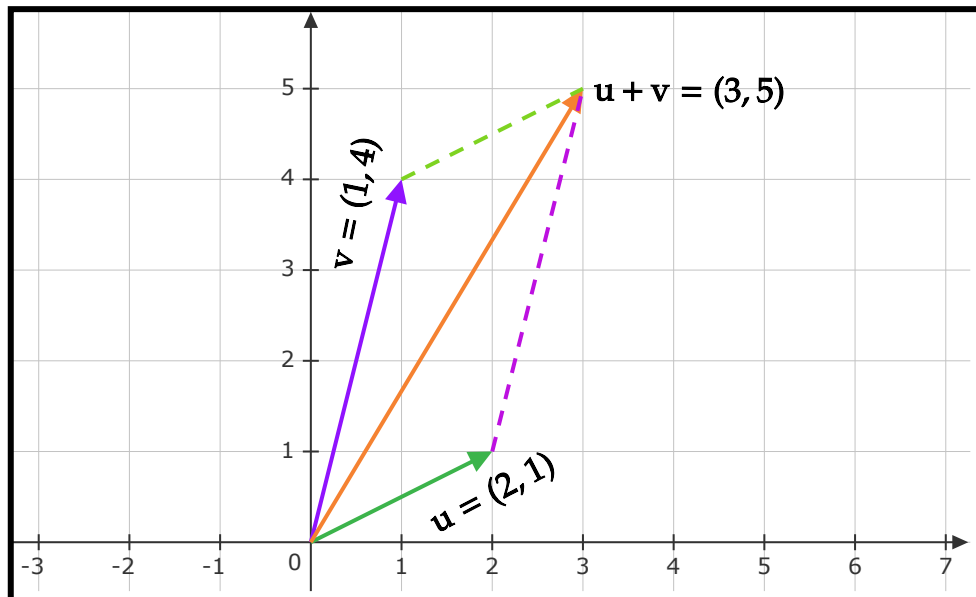


Maths 2 : Linear Algebra

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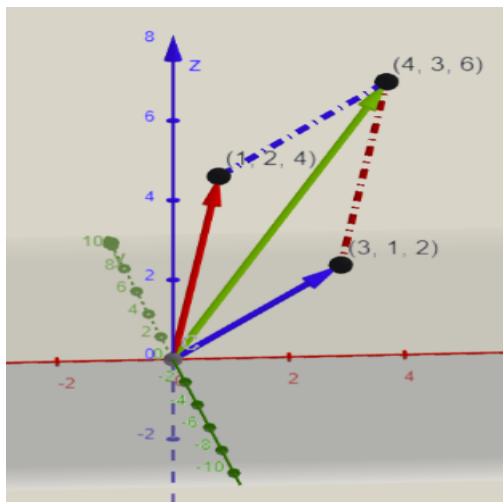
1. Parallelogram Law of Vector Addition

In $\mathbb{R}^2$

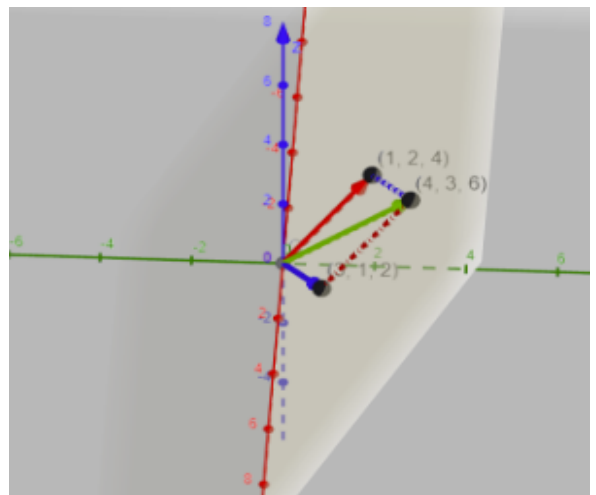


In $\mathbb{R}^3$

FRONT VIEW



TOP VIEW - All the 3 vectors are on the same plane



2. Matrix Multiplication

(Though a vector is also a matrix, for this section, by matrix, we mean a matrix with multiple rows and columns. this is just for illustration)

Matrix & Column Vector

$$\begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{bmatrix} \times \begin{bmatrix} p \\ q \\ r \\ s \end{bmatrix} = \begin{bmatrix} p(a_1) + q(b_1) + r(c_1) + s(d_1) \\ p(a_2) + q(b_2) + r(c_2) + s(d_2) \\ p(a_3) + q(b_3) + r(c_3) + s(d_3) \end{bmatrix}$$

$$= p \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + q \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} + r \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} + s \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix}$$

Multiplying a matrix (on left) and a column vector (on right), results in a column vector, which is a linear combination of the **columns** of the matrix.

Row Vector & Matrix

($\cdot$) $\rightarrow$ dot represents multiplication of a vector with a scalar

$$\begin{bmatrix} p & q & r & s \end{bmatrix} \times \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \\ d_1 & d_2 & d_3 \end{bmatrix}$$

$$=$$

$$\begin{bmatrix} p(a_1) + q(b_1) + r(c_1) + s(d_1) & p(a_2) + q(b_2) + r(c_2) + s(d_2) & p(a_3) + q(b_3) + r(c_3) + s(d_3) \end{bmatrix}$$

=

$$\begin{aligned} & p \cdot \begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} \\ & + \\ & q \cdot \begin{bmatrix} b_1 & b_2 & b_3 \end{bmatrix} \\ & + \\ & r \cdot \begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix} \\ & + \\ & s \cdot \begin{bmatrix} d_1 & d_2 & d_3 \end{bmatrix} \end{aligned}$$

Multiplying a row vector (on left) and a matrix (on right), results in a row vector, which is a linear combination of the **row** of the matrix.

3. Properties of Determinant

Let A and B two $n \times n$ matrices.

- Determinant of an identity matrix is 1
- $\det(A) = \det(A^T)$

- $\det(AB) = \det(A) \times \det(B)$
 - $\det(AB) = \det(BA)$
 - $\det(kA) = k^n \det(A)$
 - If there is a column/row **without** any non-zero entry, then the determinant of that matrix is 0
-

4. Matrix Representation of a System of Linear Equations

A **system of linear equations** is a collection of two or more linear equations involving the same variables. The following is a system of linear equations with m equations and n unknown variables.

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned}$$

The matrix representation of the above system of linear equations is $Ax = b$, where

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

- Matrix A is referred to as the **coefficient matrix**.
 - If $b = 0$, then the $Ax = b$ is a **homogeneous** system of linear equations.
-

5. Solution for a System of Linear Equations

For a system of linear equations $Ax = b$, x is the unknown vector. If a vector v satisfies $Av = b$, then v is a solution for $Ax = b$.

A system of linear equations may have

1. a unique solution
2. infinitely many solutions
3. no solution

→ A system of linear equations is **consistent** if it either has a unique solution or infinitely many solutions.

→ A system of linear equations is **inconsistent** if it does not have any solution.

A homogeneous system of linear equations is always consistent. Every matrix A will have a column vector v such that $Av = 0$.

6. Solution for a System of Linear Equations : Cramer's Rule

NOTE : If the number of equations and the number of unknowns are not **equal** then this method cannot be used.

Let $Ax = b$ be a system of linear equations with 3 equations and 3 variables, where

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad x = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

If $\det(A) = 0$, then $Ax = b$ might have infinitely many solutions or it might not have any solution. We will not be able to use the method below in that case.

If $\det(A) \neq 0$, then $Ax = b$ will have a unique solution. The procedure for finding that unique solution is given below.

Let us define A_x , A_y and A_z as follows :

$$A_x = \begin{bmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{bmatrix}, \quad A_y = \begin{bmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{bmatrix}, \quad A_z = \begin{bmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{bmatrix}$$

x, y and z are given as follows

$$x = \frac{\det(A_x)}{\det(A)} \qquad y = \frac{\det(A_y)}{\det(A)} \qquad z = \frac{\det(A_z)}{\det(A)}$$

A similar approach can be followed when there are n equations and n variables.

7. Solution for a System of Linear Equations : Using Invertible Coefficient Matrix

NOTE: If the number of equations and the number of unknowns are not equal then this method cannot be used.

Let $Ax = b$ be a system of linear equations with n equations and n variables, where $n \geq 1$.

If $\det(A) = 0$, then $Ax = b$ might have infinitely many solutions or it might not have any solution. We will not be able to use the method below in that case.

If $\det(A) \neq 0$, i.e., if A is an invertible matrix, then $Ax = b$ will have a unique solution.

When $\det(A) \neq 0$, then solution for $Ax = b$ is :

$$x = A^{-1}b$$

8. Elementary Row and Column Operations

Let A be an $m \times n$ matrix.

The possible row and column operations which can be performed on A are given in the 1st column of the table below.

Let B be the matrix we get on applying one of the row/column operations. When $m = n$, these operations might have an impact on the determinant of B . These impacts are given in column 2

Elementary Row and Column Operations	Impact on $\det(A)$ if determinant exists, i.e, if $m = n$
Switching two rows or columns.	$\det(A) = -\det(B)$
Adding a multiple of one row/column to another (not to the same row) row/column respectively.	$\det(A) = \det(B)$. - No change.
Multiplying a row/column by a scalar $t \neq 0$	$\det(A) = \frac{\det(B)}{t}$

DEFINITION - LEADING ENTRY

The first non-zero entry of a row (from left to right) is the **leading entry** of that row.

9. Row Echelon Form

Let M be an $m \times n$ matrix and let $i \in \mathbb{N}$ & $j \in \mathbb{N}$, where $1 < i \leq m$ and $1 < j \leq n$.

M is in row echelon form, if the following conditions are satisfied.

1. For all the rows with atleast one non-zero entry, the leading entry must be one.
 2. If i^{th} row has a leading entry in j^{th} column, then the leading entries of all the rows above this row should be to the left of the j^{th} column.
 3. All the rows **without** a non-zero entry should be below the rows with atleast one non-zero entry.
-

10. Reduced Row Echelon Form

Let M be an $m \times n$ matrix and let $j \in \mathbb{N}$, where $1 \leq j \leq n$.

M is in reduced row echelon form, if the following conditions are satisfied.

1. M should be in row echelon form
 2. If the j^{th} column has a leading entry, then that leading entry should be the only non-zero entry in that column.
-

11. Solution for a System of Linear Equations : Gaussian Elimination Method

Let $Ax = b$ be a system of linear equations, where

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

Step-1

Augment A and b

$$(A|b) = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right]$$

Step-2

Find the reduced row echelon form for $(A|b)$ using elementary row operations.

Step-3

Let $(R|p)$ be the reduced row echelon form of $(A|b)$

If there **exists** a row in $(R|p)$ of the form :

$$[0 \ 0 \ \dots \ 0 \mid k]$$

where $k \neq 0$, then $Ax = b$ does **not** have a solution.

If $Ax = b$ is **consistent** (has either unique or infinitely many solutions), then continue with the remaining steps.

Step-4

Let $j \in \mathbb{N}$, where $1 \leq j \leq n$.

- If the j^{th} column of R in $(R|p)$ has a leading entry of some row, then j^{th} variable is a **dependent variable**.
- If the j^{th} column of R in $(R|p)$ does not have a leading entry of any row, then j^{th} variable is an **independent variable**.
- If there are **no independent variables**, then the system has a **unique solution**.
- If there is **atleast one independent variable**, then the system has **infinitely many solutions**.

Go to **Step-5** if there is a unique solution. Go to **Step-6** if there are infinitely many solutions

Step-5 (Unique solution)

Convert $(R|p)$ into linear equations. The solution obtained from those equations is the solution for $Ax = b$.

example

$$(R|p) = \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 8 \\ 0 & 0 & 1 & 0 & 7 \\ 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Convert to equations.

$$x_1 = 0$$

$$x_2 = 8$$

$$x_3 = 7$$

$$x_4 = 5$$

Step-6 (Infinitely Many Solutions)

Let us solve through an example

$$(R|p) = \left[\begin{array}{ccccc|c} 1 & 3 & 0 & 0 & 5 & 0 \\ 0 & 0 & 1 & 0 & 7 & 8 \\ 0 & 0 & 0 & 1 & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$x_1, x_3, x_4 \rightarrow$ dependent variables

$x_2, x_5 \rightarrow$ independent variables

Convert $(R|p)$ into linear equations. Express the dependent variables in terms of the independent variables.

$$x_1 + 3x_2 + 5x_5 = 0$$

$$x_1 = -3x_2 - 5x_5 \quad (1)$$

$$x_3 + 7x_5 = 8$$

$$x_3 = 8 - 7x_5 \quad (2)$$

$$x_4 = 7 \quad (3)$$

The following set is the solution set for $Ax = b$

$$\{(-3x_2 - 5x_5, \ x_2, \ 8 - 7x_5, \ 7, \ x_5) \mid x_2, x_5 \in \mathbb{R}\}$$

12. Vector Space

A **non-empty** set V is a vector space with two operations

- Addition $(+)$: $V \times V \rightarrow V$
- Scalar Multiplication $(\cdot)$: $\mathbb{R} \times V \rightarrow V$

which satisfies the following conditions for all vectors $v, u, w \in V$ and for all scalars $a, b \in \mathbb{R}$.

1. **Additive Closure** : $v + u \in V$
2. **Multiplicative Closure** : $a \cdot v \in V$
3. **Additive Commutativity** : $u + v = v + u$
4. **Additive Associativity** : $(u + v) + w = u + (v + w)$
5. **Existence of zero vector** : There should exist a zero vector v_0 in V such that $v + v_0 = v$ for all $v \in V$
6. **Additive Inverse** : For every $v \in V$, there should exist a $v' \in V$ such that $v + v' = v_0$
7. **Unity** : $1 \cdot u = u$ for $u \in V$
8. **Associativity of Multiplication** : $(ab) \cdot u = a \cdot (b \cdot u)$
9. **Distributivity** : $(a + b) \cdot u = a \cdot u + b \cdot u$ and $a \cdot (u + v) = a \cdot u + a \cdot v$

A vector is an element of a vector space.

Remember : Even if one of the vector space axiom fails to hold, then V not a vector space.

Zero vector is not always $(0, 0, \dots, 0)$. It depends of the definition of addition and scalar multiplication. Example $\rightarrow$ Linear Algebra Book (Maths-2), Example 3.2.6, Page Number 105

The set of all the solutions for a homogeneous system of linear equations is a vector space.

13. Properties of Vector Space

NOTE : Let V be a vector space. Here, $(\cdot)$ represents the scalar multiplication defined for the vector space V . Also, 0 refers to the zero scalar and v_0 refers to the zero vector of V .

1. Zero vector is unique.
2. For each $u \in V$, the additive inverse u' is unique.
3. **Cancellation Law** : If $\mathbf{v}_1 + \mathbf{v}_3 = \mathbf{v}_2 + \mathbf{v}_3$ for $v_1, v_2, v_3 \in V$, then $\mathbf{v}_1 = \mathbf{v}_2$.
4. **Cancellation Law** : If $\mathbf{v}_3 + \mathbf{v}_1 = \mathbf{v}_3 + \mathbf{v}_2$ for $v_1, v_2, v_3 \in V$, then $\mathbf{v}_1 = \mathbf{v}_2$.
5. $0 \cdot \mathbf{v} = \mathbf{v}_0$ for all $v \in V$. Here
6. $c \cdot \mathbf{v}_0 = \mathbf{v}_0$ for all $v \in V$. Here $c \in \mathbb{R}$.
7. For each $c \in \mathbb{R}$ and for each $u \in V$, we have $(-c) \cdot \mathbf{u} = -(c \cdot \mathbf{u}) = c \cdot (-\mathbf{u})$.
8. If $\mathbf{Ax} = \mathbf{b}$ and $\mathbf{b} = \mathbf{0}$ (a homogeneous system), then the solution set is a Vector Space.

14. Subspace of a Vector Space

Let V be a vector space. Let U be a **non-empty** subset of V . U is a subspace of V if and only if the following conditions are true.

1. If $u_1, u_2 \in U$, then $u_1 + u_2$ should also be in U .
2. If $u \in U$, then $c \cdot u$ should also be in U . Here, $c \in \mathbb{R}$.

U is a **proper subspace** of V if U is a subspace and also an proper subset of V .

Two trivial subspaces of a vector space V

- V is a subspace of V
- The subset $\{v_0\}$ is a subspace of V . Here, v_0 is the zero vector of V .

15. Linear Combination

Let V be a vector space. Let $v_1, v_2, \dots, v_n \in V$ and let $a_1, a_2, \dots, a_n \in \mathbb{R}$.

$$z = a_1 v_1 + a_2 v_2 + \dots + a_n v_n = \sum_{i=1}^n a_i \cdot v_i$$

z is a **linear combination** of $v_1, v_2, \dots, v_n \in V$.

16. Linear independence and Linear dependence.

Let $S = \{v_1, v_2, \dots, v_n\}$ be a set of vectors in $\mathbb{R}^d$ where $d \in \mathbb{R}$.

Consider the following system

$$\begin{bmatrix} | & | & \cdots & | \\ v_1 & v_2 & & v_n \\ | & | & & | \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

- If this system has a **unique solution**, then S is a **linearly independent** set.
- If this system has **infinitely many solutions**, the S is a **linearly dependent** set.

In simple words, if there exists **only** one way to linearly combine the vectors in set S to get $[0 \ 0 \ \dots \ 0]^T$, then S is **linearly independent**. If there are more than one way to linearly combine the vectors in set S to get $[0 \ 0 \ \dots \ 0]^T$, then S is **linearly dependent**.

NOTE : If set S is linearly independent, then $[x_1 \ x_2 \ \dots \ x_n]^T = [0 \ 0 \ \dots \ 0]^T$. This is because, $[0 \ 0 \ \dots \ 0]^T$ is always a solution to a homogeneous system of linear equations.

17. Span Set

Let $S = \{v_1, v_2, \dots, v_n\}$ be a set of vectors in $\mathbb{R}^d$ where $d \in \mathbb{R}$.

The span set of S is defined as follows.

$$\text{Span}(S) = \left\{ \sum_{i=1}^n a_i \cdot v_i \mid v_i \in S \text{ and } a_i \in \mathbb{R} \right\}$$

In simple words, $\text{span}(S)$ is the set of all possible linear combinations of $v_1, v_2, \dots, v_n$.

NOTE : $\text{Span}(S)$ is a vector space.

18. Basis and Dimension of a Vector Space

Let V be a vector space. Let B be a subset of V . B is a basis of V if it satisfies the following conditions.

- B is a linearly independent set
- $\text{Span}(B) = V$

If B is a basis of V , then the **dimension** of V is the cardinality of B . In simple words, the **dimension** of V is the number of elements in the basis of V .

The dimension of a vector space is unique. The basis set is not unique.

A set S is a **maximal linearly independent** set if S is a linearly independent set any super set of S is a linearly dependent set.

A set S is a **minimal spanning set** if $\text{Span}(S) = V$ and any proper subset of S does not span V .

The following conditions are **equivalent** for a set B to be a basis of a vector space V :

1. B is linearly independent and $\text{span}(B) = V$.
 2. B is a maximal linearly independent set.
 3. B is a minimal spanning set.
-

19. Finding a Basis Set

Let $S = \{v_1, v_2, \dots, v_n\}$ be a set of vectors.

We have to find the basis of $\text{Span}(S)$

Method 1

- Let matrix A be the following.

$$A = \begin{bmatrix} \text{---} & v_1 & \text{---} \\ \text{---} & v_2 & \text{---} \\ & \vdots & \\ \text{---} & v_n & \text{---} \end{bmatrix}$$

- Let R be the reduced row echelon form of A .
- Let B be the set of all the rows R with atleast one non-zero entry.

B is a basis set for $\text{Span}(S)$

Method 2

- Let matrix A be the following.

$$A = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{bmatrix}$$

- Let R be the reduced row echelon form of A .
- Let $B = \{\}$ be an empty set.
- If the i^{th} column of R has a leading entry of a row, then append the i^{th} column vector A to the set B .
- Check this for every column in R .

In then end, set B will be a basis for $\text{Span}(S)$

20. Rank of a Matrix

Let A be an $m \times n$ matrix.

The rank of A is the number of linearly independent columns of A

- Column space of $A \rightarrow \text{Span}\{\text{All the column vectors of } A\}$
- Column rank of $A \rightarrow \dim(\text{Column space of } A)$.
- Row space of $A \rightarrow \text{Span}\{\text{All the row vecors of } A\}$
- Row rank of $A \rightarrow \dim(\text{Row space of } A)$

$$\text{Row Rank}(A) = \text{Column Rank}(A)$$

21. Nullity of a Matrix

Let A be an $m \times n$ matrix.

The set of all the solutions for $Ax = 0$ is the nullspace of A .

$$\text{Nullspace}(A) = \{x | Ax = 0\}$$

The dimension of the $\text{Nullspace}(A)$ is the nullity of A .

22. Rank-Nullity Theorem of Matrices

Let A be an $m \times n$ matrix. Then

$$\text{Rank}(A) + \text{Nullity}(A) = n$$

23. Dot Product

Let $u = (u_1, u_2, \dots, u_n)$ and let $v = (v_1, v_2, \dots, v_n)$, where $u, v \in \mathbb{R}^n$

The dot product between two u and v is defined as

$$u \cdot v = (u_1 \times v_1) + (u_2 \times v_2) + \dots + (u_n \times v_n)$$

If u and v are column vectors, the $u \cdot v$ can be written as $u^T v$.

Let $c \in \mathbb{R}$. Some properties of dot product are as follows :

- $u \cdot v = v \cdot u$
 - $cv \cdot u = c(v \cdot u)$
 - $cu \cdot cv = c^2(u \cdot v)$
-

24. Length of a Vector

Let $v = (a_1, a_2, \dots, a_n)$. The length of v is given by

$$\text{Length}(v) = \sqrt{v \cdot v} = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}$$

25. Euclidean Distance Between Two Vectors

Let $v, u \in \mathbb{R}^n$. The Euclidean distance between the two vectors is given by:

$$||v - u|| = \sqrt{\sum_{i=1}^n (v_i - u_i)^2}$$

26. Unit Vector

v is a **unit vector** if the length of v is equal to 1.

Let k be a vector in $\mathbb{R}^n$.

The unit vector in the direction of k is $\Rightarrow \boxed{\frac{k}{Length(k)}}$

27. Angle Between Two Vectors

Let $u = (u_1, u_2, \dots, u_n)$ and let $v = (v_1, v_2, \dots, v_n)$, where $u, v \in \mathbb{R}^n$

Let θ° be the angle between u and v . Then,

$$\theta = \cos^{-1} \left(\left(\frac{u}{Length(u)} \right) \cdot \left(\frac{v}{Length(v)} \right) \right) = \cos^{-1} \left(\left(\frac{u}{||u||} \cdot \frac{v}{||v||} \right) \right)$$

If the dot product between two vectors is 0, then the angle between these two vectors is 90° .

28. Orthogonal Complement

V and U are two subspaces in $\mathbb{R}^n$ such that $\dim(V) + \dim(U) = n$

If every $u_i \in U$ is orthogonal to every $v_i \in V$, then U and V are orthogonal complements
